

SEMESTER END EXAMINATIONS – MARCH 2024

Program	: B.E. – CSE (Artificial Intelligence and Machine Learning) / CSE (Cyber Security)	Semester	: III
Course Name	: Computer Organization and Architecture	Max. Marks	: 100
Course Code	: CI34 / CY34	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT – I

- Differentiate between combinational and sequential circuits. Describe the working of full adder with neat circuit diagram and truth table. CO1 (08)
 - Minimize the following expression using laws of Boolean algebra: $(A+B+C)(A+\bar{B}+C)(A+B+\bar{C})$. CO1 (04)
 - Describe the working of JK- flipflop with neat circuit diagram and truth table. CO1 (08)
- Describe the working of RS flipflop with the neat circuit diagram and truth table. CO1 (08)
 - With neat circuit diagram and truth table briefly describe 4:1 multiplexer. CO1 (08)
 - Consider a logic circuit with three inputs A, B, C. Output Y is high for the following conditions:
 - B and C are true
 - A and C are false
 - A, B, and C are true
 - A, B and C are false.

Minimize the function Y using Law's of Boolean Algebra.

UNIT – II

- Explain any Five data transfer instructions used by MIPS/x86 architecture. CO2 (06)
 - List the difference between high-level language and machine language? CO2 (06)
 - How to measure the performance of computer? Explain. CO2 (08)
- With an example discuss the relative addressing mode. CO2 (06)
 - Explain assembly directive? List the few assembly directives and which directive are used to inform the assembler. CO2 (06)
 - With an Example, explain Direct and indirection instruction set of MIPS/x86 assembly language. CO2 (08)

UNIT – III

- Explain with a neat diagram bus arbitration using Daisy chain. CO3 (08)
 - Explain with a neat diagram how the DMA controllers are used in computer system. CO3 (06)
 - Design and Implement 4 bit carry look ahead adder with neat circuit diagram. CO3 (06)

6. a) Explain the I/O interface for an input device. CO3 (06)
b) List the functions of interface circuits and describe how the Printer to processor connection is handled in a system with a neat block diagram. CO3 (08)
c) Explain Binary addition- subtraction logic network. CO3 (06)

UNIT – IV

7. a) Describe the following: CO3 (06)
i. Memory access time
ii. Memory cycle time
iii. Function of memory management unit
iv. Logical or virtual memory
v. Significance of cache memory.
b) Illustrate the working of static random access and dynamic random-access memories with their block diagrams. CO3 (06)
c) Write the algorithm of restoring division method. Perform the division on the numbers 16/4 using the same algorithm and write register circuit arrangement to perform the same. CO3 (08)
8. a) Illustrate associative and set-associative cache memory mapping techniques with an example. CO3 (06)
b) Perform the multiplication on the numbers 10 x 08 using ripple carry addition method and carry save addition method. Write the carry-save adder circuit diagram for 4-bit operands and mention reasons why CSA working faster than the ripple carry adder. CO3 (08)
c) Perform multiplication on the numbers 8 x -11 using Booth recoding and Booth bit pair coding methods. CO3 (06)

UNIT- V

9. a) List out the key aspects of GPU programs driving GPU processor architecture and explain any one in briefly. CO5 (06)
b) List out the difference of CSIC and RISC style processors. CO5 (08)
c) Discuss the basic organization of micro programmed control unit with neat circuit diagram. CO5 (06)
10. a) List the sequence of actions needed to fetch and execute the instruction Store R6, X(R8) and Load R5, X(R7). CO5 (06)
b) Illustrate along with key characteristics as how GPUs vary from CPUs. CO5 (08)
c) Explain the working of Hardwired Control Unit with a neat diagram. CO5 (06)
